

PRINCIPIA ORTHOGONA

Volume Two: Contact Realization of Generative Transitions

Version V4 — Deposit Edition (July 2026)

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GitHub: github.com/TOTOGT/AXLE · AXLE Issue #13 (Gronwall asymmetry, closed)

Lean 4: `AXLE/lean/VolumeTwo.lean`

Dedicated to my children Vic (R.I.P.), Giulia, Alice (R.I.P.), Sarah (R.I.P.), and David. Once tiny, always strong.

Abstract

This volume is the second in the *Principia Orthogona* series. Volume One developed the singularity-theoretic and variational foundations: the operator sequence $C \rightarrow K \rightarrow F \rightarrow U$, the curvature threshold κ^* , the Whitney A_1 - A_3 singularity classification, and a symplectic preservation theorem for the fold map. The present volume constructs the explicit contact-geometric realization of those foundations. The three main results are: (A) a precise correspondence between the geometric fold operator F and the dm^3 contact-Hamiltonian dissipation H_{diss} (§2); (B) equivalence of the curvature threshold κ^* and the embodiment threshold τ (§3), with explicit values $\tau = 2$, $\varepsilon_0 = 1/3$ verified in the dm^3 toy model; (C) a bifurcation analysis showing that the four dm^3 bifurcations correspond bijectively to the Whitney A_1 - A_3 singularity types (§5). Version V4 corrects: the Contact Hopf bifurcation point γ^* (§5.1), from e^{z_0} to $2e^{z_0}$, propagated from the dm^3 toy model paper's V3 erratum. No other theorem statement, canonical invariant, or Lean-verified fact changes from V3. Version V3 added: the certified inner-basin boundary $r^* = 0.77594059$ (rounded ≈ 0.776), refining the placeholder value 0.773 used in V2a; an updated Gronwall-asymmetry section (§6.3, closing AXLE Issue #13); Lean 4 formal proof skeleton (VolumeTwo.lean); seven reproducible figures (figures.py); and an interactive HTML dashboard.

MSC 2020: 37C10 (limit cycles), 53D10 (contact manifolds), 37C75 (stability), 58K05 (Whitney singularities), 37G10 (bifurcations), 70H05 (Hamiltonian systems).

Keywords: contact geometry, dm^3 toy model, Whitney singularities, Gronwall stability, Lean 4 formal verification, curvature threshold, embodiment threshold, operator algebra, generative transitions, helical attractor.

Preface

This volume is the second in the *Principia Orthogona* series. Volume One [1] developed the singularity-theoretic and variational foundations of generative transitions: the operator sequence $C \rightarrow K \rightarrow F \rightarrow U$, the curvature threshold κ^* , the Whitney A_1 - A_3 singularity classification, and a symplectic preservation theorem for the fold map.

The central results are:

- A precise correspondence between the geometric fold operator F and the dm^3 contact-Hamiltonian dissipation (§2).
- A theorem establishing the equivalence of the curvature threshold κ^* and the embodiment threshold τ (§3).
- Explicit verification of the correspondence in the dm^3 toy model (§4).
- A bifurcation analysis showing that the four dm^3 bifurcations correspond to the singularity types of Volume One (§5).

Throughout, we take as established the results of Volume One [1], Generative Contact Mechanics [2], and the dm^3 Toy Model [3]. Volume III instantiates the framework in biological systems; Volume IV delivers GTCT T1 and the IMPA work; Volume V + AXLE provides the Banach FPT formal proof and Complete Completeness.

1 Introduction

1.1 The Problem: From Geometric Folds to Dissipative Dynamics

Volume One established that generative transitions are localized geometric events classified by the Whitney A_1 - A_3 hierarchy. The fold operator F was shown to act as a symplectic canonical transformation on T^*X , and the full transition $G = U \circ F \circ K \circ C$ was shown to be a piecewise-smooth symplectic map. What Volume One deliberately left open was the question of *post-fold stability*: once the fold has occurred and the unfolding U has selected a stable branch, what governs the long-term dissipative dynamics near that branch?

The Hamiltonian framework of Volume One cannot answer this question: Liouville's theorem forbids attractors in symplectic systems on compact manifolds. The answer is contact geometry. The contact manifold $M = X \times \mathbb{R}$ with contact form $\alpha = dz - \lambda$ provides the correct geometric setting for dissipative dynamics with limit-cycle attractors, stochastic stability, and variational structure.

1.2 Role of the dm^3 Toy Model

The dm^3 Toy Model [3] serves as a complete, explicit realization of the abstract contact-geometric framework of [2] and as the bridge that certifies the realizability of the geometric fold theory of Volume One. The dm^3 Toy Model proves that the generative transition framework is not merely definable, but fully instantiable by an explicit, smooth, globally analyzable dynamical system. Its role is logical, not empirical.

1.3 Main Results

Theorem 1.1 (Contact Realization of the Fold). *The fold operator F of Volume One is the piecewise-smooth, pre-contact limit of the dm^3 operator $A_{dm^3} = \varphi^{T^*/4}$ of [2]. Under the contact extension $M = X \times \mathbb{R}$, the impulsive momentum jump $p^+ - p^- = \mu n$ at the fold corresponds to the contact-Hamiltonian correction $H_{\text{diss}} = -\gamma V e^{-\beta z}$ in the regularized limit $\beta \rightarrow \infty$.*

Theorem 1.2 (Threshold Equivalence). *$|\kappa| \uparrow \kappa^* \iff \mu_{\max} < 0 \iff \tau = \sqrt{c/\kappa_{\text{noise}}} \in (0, \infty)$. The curvature threshold κ^* and the embodiment threshold τ are two parameterizations of the same event: the onset of transverse stability in the post-fold dissipative system.*

Theorem 1.3 (Singularity–Bifurcation Correspondence). *The four bifurcations of the dm^3 toy model [3] correspond to the Whitney singularity types of Volume One under the projection $M = S \times \mathbb{R} \rightarrow S$.*

1.4 Standing Assumptions

Assumption 1.4 (Volume One framework). *The operator sequence $C \rightarrow K \rightarrow F \rightarrow U$, the threshold κ^* , and all results of [1] hold.*

Assumption 1.5 (dm^3 framework). *The dm^3 system, contact manifold $M = S \times \mathbb{R}$, and all results of [2, 3] hold.*

2 Contact Realization of the Fold Operator

2.1 From Hamiltonian Phase Space to Contact Extension

In Volume One, the fold is realized in T^*X as a symplectic discontinuity. At fold point s_0 :

$$p(s_0^+) - p(s_0^-) = \mu n(s_0)$$

generated by $S(\gamma) = \mu \Theta(\kappa(\gamma) - \kappa^*)$ ([1], §12). The fold map $F : (\gamma, p) \mapsto (\gamma, p + \mu n)$ preserves $\omega = d\gamma \wedge dp$ ([1], Theorem 11.1). To capture post-fold stabilization, we pass to the contact extension $M = X \times \mathbb{R}$ with $\alpha = dz - \lambda$, $d\lambda = \omega$ ([2], Def. 6.1).

2.2 The Fold as a Contact Discontinuity

Proposition 2.1 (Regularization of the fold generator). *The contact dissipation $H_{\text{diss}}(x, z) = -\gamma V(x) e^{-\beta z}$ is a smooth regularization of $S(\gamma) = \mu \Theta(\kappa(\gamma) - \kappa^*)$: as $\beta \rightarrow \infty$ and $z \rightarrow 0^+$, the correction $-\gamma \nabla V e^{-\beta z}$ concentrates near $\Gamma = \{V = 0\}$ and mimics the threshold activation of Θ at $\kappa = \kappa^*$. Structural properties: (i) $H_{\text{diss}}|_{\Gamma} = 0$; (ii) off Γ : $H_{\text{diss}} < 0$; (iii) $e^{-\beta z}$ weakens dissipation as action accumulates.*

2.3 Relation to the dm^3 Normal Form

By [2] (Theorem C), every dm^3 system near Γ is locally contact-diffeomorphic to:

$$\begin{aligned} \dot{\rho} &= \mu_{\max}(1 - e^{-\beta z})\rho + O(\rho^2) \quad (\text{radial}) \\ \dot{\theta} &= \omega + O(\rho) \\ \dot{z} &= \omega - |\mu_{\max}|\rho^2 e^{-\beta z} + O(\rho^3) \end{aligned}$$

2.4 Correspondence Table (proves Theorem A)

Volume One (geometric)	GCM / dm^3 (contact)
Curvature threshold κ^*	Onset of transverse contraction
Fold impulse $p^+ - p^- = \mu n$	Contact correction H_{diss}
Rank-1 Jacobian loss	Dissipative contact normal form
Unfolding U	Gradient flow to Γ via U_3 [2]
Distributional generator S	Regularized H_{diss} (Prop. 2.1)

3 Equivalence of κ^* and τ

3.1 Geometric Threshold Implies Stochastic Threshold

Lemma 3.1 (Fold activation produces hyperbolicity). *If K drives curvature to κ^* , F is rank-1, and U selects a nondegenerate branch, then Γ is hyperbolic with $\mu_{\max} < 0$ and $\dot{V} \leq -cV$ for some $c > 0$.*

Theorem 3.2 (Geometric implies stochastic threshold). *Under the previous lemma, $\mathcal{L}V \leq -cV + \kappa_{\text{noise}}\|\sigma\|^2$ and $\tau = \sqrt{c/\kappa_{\text{noise}}} \in (0, \infty)$.*

3.2 Stochastic Threshold Implies Geometric Threshold

Lemma 3.3 (Finite τ implies transverse contraction). *If $\mathcal{L}V \leq -cV + \kappa_{\text{noise}}\|\sigma\|^2$ with $c > 0$, then $\mu_{\max} < 0$ and $\dot{V} \leq -cV$ near Γ .*

Theorem 3.4 (Stochastic implies geometric threshold). *If $\tau \in (0, \infty)$, then the trajectory must have crossed κ^* and undergone a rank-1 fold.*

3.3 The Equivalence Theorem

Theorem 3.5 (Threshold Equivalence — Theorem B). $|\kappa| \uparrow \kappa^* \iff \mu_{\max} < 0 \iff \tau \in (0, \infty)$. *The threshold κ^* is the geometric precursor of τ : curvature accumulation creates the conditions under which stochastic stability becomes meaningful. Scope: local to the fold neighborhood and post-fold tubular neighborhood of Γ .*

4 Explicit Verification in the dm^3 Toy Model

4.1 The System and Verification Theorem

Theorem 4.1 (Verification of GCM by the dm^3 Toy Model). *There exists an explicit smooth dynamical system on a contact manifold — the dm^3 toy model — in which all of the following hold by direct computation: (1) Axiom consistency: all eight dm^3 axioms satisfied simultaneously; (2) Contact structure: explicit $\alpha = dz - r^2 d\theta$; (3) Normal form: contact-diffeomorphic to universal form with $(\mu_{\max}, \omega, \beta) = (-2, 1, 1)$; (4) Operator algebra closure; (5) Stochastic stability: $\tau = 2$ in closed form, Gaussian stationary measure; (6) Global dynamics: attractor Γ_{12} , all four predicted bifurcations.*

4.2 The System

On $M = \mathbb{R}_{>0}^2 \times \mathbb{R}$ with polar coordinates (r, θ, z) and contact form $\alpha = dz - r^2 d\theta$ (Proposition 2.2 of [3]):

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z} \quad (1)$$

$$\dot{\theta} = 1 \quad (2)$$

$$\dot{z} = r^2 - 2(r - 1)^2 e^{-z} \quad (3)$$

Limit cycle: $\Gamma = \{r = 1\}$, $T^* = 2\pi$. These are the exact equations used in `figures.py` (fully reproducible: `python3 figures.py`).

4.3 Threshold Verification

Proposition 4.2 (Explicit threshold values). $\mu_{\max} = -2$, $\kappa_{\text{noise}} = 1$, $\tau = 2$. *The transverse eigenvalue $\lambda(z) = -2(1 - e^{-z})$ satisfies: $\lambda(0) = 0$ (neutral, pre-embodiment); $\lambda(z) < 0$ for $z > 0$ (attracting, post-embodiment); $\lambda(z) \rightarrow -2$ as $z \rightarrow \infty$.*

The neutral stability at $z = 0$ is the mathematical content of the embodiment threshold: the orbit earns its stability by accumulating action. The crossing $z = 0 \rightarrow z > 0$ corresponds exactly to the curvature crossing $\kappa \rightarrow \kappa^*$ in Volume One.

4.4 Normal Form Verification

Proposition 4.3 (Contact normal form). *In coordinates (ρ, θ, z) with $\rho = r - 1$, the system takes the form $\dot{\rho} = -2(1 - e^{-z})\rho + O(\rho^2)$, $\dot{\theta} = 1 + O(\rho)$, $\dot{z} = 1 - 2\rho^2 e^{-z} + O(\rho^3)$, the contact normal form of [2] (Theorem C) with $(\mu_{\max}, \omega, \beta) = (-2, 1, 1)$.*

4.5 Stability Radius (outer basin)

$$\varepsilon_0 = \frac{|\mu_{\max}|}{2(1 + \sup_{\Gamma} \|\text{Hess } V\|)} = \frac{2}{2(1 + 2)} = \frac{1}{3}.$$

4.6 Inner-basin boundary r^* (V3 correction, AXLE Issue #13)

The symmetric Gronwall estimate $|r - 1| < 1/3$ correctly bounds the *outer* basin $\{r > r_{\text{att}}\}$, but the *inner* basin is asymmetric: trajectories starting at $r_0 \in (2/3, r^*)$ lie inside the Gronwall ball yet still escape to $r \rightarrow 0$. The true inner boundary, certified numerically by `certify_rstar.py` (DOP853, `rtol` = 10^{-12} , `atol` = 10^{-14} , bisection tolerance 10^{-7}), is

$$r^* = 0.77594059 \text{ (rounded to 3 d.p.: 0.776)}.$$

The corrected hierarchy is

$$\varepsilon_0 = \frac{1}{3} < \frac{2}{3} < r^* \approx 0.776 < \kappa^* = \sqrt{7/9} \approx 0.882 < 1.$$

This refines the placeholder value $r^* \approx 0.773$ used in V2a (the deviation, $\approx 3 \times 10^{-3}$, is the bisection-floor of the prior certification run; the present value is reproducible on any IEEE-754 platform to ≥ 6 significant figures). AXLE Issue #13 is hereby closed; the Lean obligation `thm_gronwall_asymmetry` now has a numerical certificate.

5 Singularity-Bifurcation Correspondence

5.1 The Four Bifurcations of the Toy Model

From [3] (Theorem C), the dm^3 toy model exhibits four bifurcations:

- (i) Contact Hopf at $\gamma = 2e^{z_0}$: limit cycle loses stability, new cycle bifurcates. (Corrected in V4: the dm^3 toy model paper’s V3 erratum found a spurious factor of 2 in the linearized transverse eigenvalue, giving $\gamma^* = e^{z_0}$ in earlier versions where the correct value is $\gamma^* = 2e^{z_0}$; confirmed there by an internal consistency check against the paper’s own baseline parameter $\gamma = 2$.)
- (ii) Saddle-node at $\eta \approx 0.15$: two cycles collide, multi-orbit structure collapses.
- (iii) Neimark-Sacker at detuning $|\Delta| = \Delta^*$: resonant orbit loses stability, 2-torus bifurcates.
- (iv) Slow-fast crossover at $\beta = \beta^*$: smooth transition between contact and classical regimes.

5.2 Correspondence with Whitney Singularities

Proposition 5.1 (Singularity-Bifurcation Correspondence). *Under projection $M = S \times \mathbb{R} \rightarrow S$: A_1 (codim 0) $\leftrightarrow$ Contact Hopf + Saddle-node; A_2 (codim 1) $\leftrightarrow$ Neimark-Sacker; A_3 (codim 2) $\leftrightarrow$ Slow-fast crossover. Higher singularities are excluded: in Volume One by the Morse condition; in [2] by contact normal form rigidity (Theorem C). Proves Theorem C.*

Whitney	Codim	dm^3 Bifurcation	Mechanism
A_1 (fold)	0	Contact Hopf (H)	Rank-1 loss, radial direction
A_1 (fold)	0	Saddle-node (SN)	Rank-1 loss, radial collision
A_2 (cusp)	1	Neimark-Sacker (NS)	Rank-1, 2nd angular direction
A_3 (swallowtail)	2	Slow-fast (SF)	Rank-1, 3rd-order z -change

6 Discussion

6.1 Summary of the Three Main Results

1. The fold operator F is the geometric precursor of the dm^3 contact-Hamiltonian H_{diss} ; the distributional generator S is regularized by H_{diss} in the limiting sense of Proposition 2.1 (Theorem A).
2. κ^* and τ are equivalent: each is finite and positive iff the other is, both detecting the onset of transverse stability (Theorem B).
3. The four dm^3 bifurcations correspond to Whitney A_1 - A_3 types, completing the singularity-theoretic classification of the toy model dynamics (Theorem C).

6.2 Role in the G-Series (G^1 - G^5)

The *Principia Orthogona* series is a spiral, not a ladder. Each volume is a complete orbit — a full turn around the fixed point. G applied five times is $G^5 = \text{Complete Completeness}$. G^6 remains open (2026).

<i>G</i> -Level / Volume	Core Concept	Fixed Point
G^1 / Vol. I	Abstract Operator Algebra	Orthogonal operator
G^2 / Vol. II (this)	Contact Geometry, $g_{33} = 33$	Contact fixed point
G^3 / Vol. III	Biological Instantiations	Living form
G^4 / Vol. IV	GTCT T1, IMPA	Temporal contact
G^5 / Vol. V + AXLE	Banach FPT, formal proof	Complete Completeness
G^6 / Issue 6	$\chi(H^*(X^6)) = 33 \forall n$	Open — 2026

6.3 Open Problems

1. **Global Equivalence.** Theorem B is local. A global version requires showing every τ -stable dm^3 system arises from a fold globally on X .
2. **Higher Resonances.** Systematic treatment of $k:m$ correspondence between higher A_k singularities and higher resonances.
3. **Volume Three.** Domain-specific (X, g, Φ) with explicit computation of κ^* and τ from data.
4. **Gronwall Asymmetry (AXLE Issue #13, closed in V3).** The numerical certification $r^* = 0.77594059$ (§4.7) establishes the inner-basin boundary. A fully formal Lean 4 proof remains open (thm_gronwall_asymmetry, ***).

Appendix A — Lean 4 Formal Verification Status (V3)

All theorems have corresponding obligations in AXLE/lean/VolumeTwo.lean.

Theorem / Lemma	Lean Name	Status
$\lambda(0) = 0$ (Prop. 4.2)	eigenvalue_at_zero	proved
$\lambda(z) < 0$ for $z > 0$ (Prop. 4.2)	eigenvalue_neg_pos_z	proved
$\tau > 0$ (Thm 3.2)	embodimentThreshold_pos	proved
$\tau = 2$ (Prop. 4.2)	toyModel_tau	proved
$\varepsilon_0 = 1/3$ (§4.6)	toyModel_epsilon0	proved
Thm C bijection (Prop. 5.1)	thm_C_singularity_bijection	proved
$\mu_{\max} < 0 \iff \tau > 0$	thm_B_mu_iff_tau	proved
A_1 surjectivity	thm_C_A1_surjective	proved
$\lambda(z) \rightarrow \mu_{\max}$ (filter)	eigenvalue_limit_filter	sorry **
Thm A (full)	thm_A_contact_realization	sorry ****
Thm B (full chain)	thm_B_full_chain	sorry *****
Gronwall asymmetry	thm_gronwall_asymmetry	sorry *** (numeric cert. V3)

Policy. No sorry is hidden. Every open obligation is documented with a proof strategy and difficulty rating (* = easy, ***** = frontier). File: github.com/TOTOGT/AXLE/lean/VolumeTwo.lean.

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Version history

- **V4 (July 2026, DOI 10.5281/zenodo.21148424):** corrected the Contact Hopf bifurcation point $\gamma^* = e^{z_0} \rightarrow 2e^{z_0}$ (§5.1), propagated from the dm^3 toy model V3 erratum; no other change.
- **V3 (June 2026, DOI 10.5281/zenodo.20755436):** certified inner-basin boundary $r^* = 0.77594059$ (§4.7); AXLE Issue #13 closed; updated Lean status table; cosmetic edits.
- **V2a (May 2026, DOI 10.5281/zenodo.20159456):** Lean 4 skeleton; 7 reproducible figures; HTML dashboard; placeholder $r^* \approx 0.773$.

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